

Elastic: Redesigning Habit Tracking

Prototype

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Most habit trackers are built around the assumption that users show up with consistent energy. They don't. Elastic is a habit tracking system designed around that reality – capturing partial effort, reducing logging friction, and replacing punitive streaks with a forgiving momentum model. The result is a system where a low-energy day is never a reason to quit.

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1. The Trigger (Why I Built This)

At the beginning of 2026, I tried to become more consistent with habits like running, coding, and reading. I experimented with many tools, mainly notebooks, Notion, and Habitica— each of them solved part of the problem but failed as a complete system.

- Notebooks had low friction but no insights
- Notion was flexible but required repetitive manual logging, turning tracking into a chore
- Habitica was engaging due to its gamification but heavy and unreliable, creating friction in onboarding and daily use

Despite trying all three, I kept failing. The assumption was that this reflected a lack of discipline. The more useful observation was that none of the systems could survive a low-energy day – and low-energy days are not exceptions, they are the norm.

Core Insight: Habit trackers fail not because users lack discipline, but because the systems assume consistent energy.

Most habit tools (such as Duolingo, Wordle, and Snapchat) operate on a **binary “all or nothing” model**. Users either complete a task or fail it. This creates a fragile system where missing a single day, or falling short of the ideal effort, feels like a complete collapse of progress.

While platforms like Strava offer more "elasticity" by focusing on weekly goals rather than daily streaks, they still suffer from **flattened effort**. They treat a 30-minute recovery walk and a 3-hour intense run as identical "active days". This ignores the true quality and intensity of work, making the tracked progress feel disconnected from the user's reality.

My prototype, **Elastic**, bridges this gap through a Tier System (Mini, Plus, Elite). By capturing the specific intensity of each session:

- Low-energy days are preserved through “Mini” habits: tasks that are too easy to fail
- High-energy days are rewarded as “Elite” sessions: encouraging users to achieve beyond their set goals

Why this helps Retention:

The result of traditional tracking is predictable: users disengage when the **system fails to recognize partial effort**, not when users fail completely. On low-energy days, users often believe they require more discipline when in reality, they need a system that allows them to **scale down without dropping off**. This shift from “perfect execution” to “continuous forward movement” is the foundation of Elastic.

2. Research & Validation

To validate whether the problem was personal or systemic, I surveyed 16 habit tracker users and conducted 5 follow-up interviews across different habit types: fitness, studying, and creative work.

Key findings:

- 75% reported abandoning a habit after a broken streak despite still intending to continue
- 88% said partial effort counted as failure in their current tracker
- 69% had stopped tracking at least once because logging felt like too much effort
- 81% reported no visibility into whether their consistency patterns were improving or declining

Two findings shifted the product direction:

First, users weren't quitting because they stopped caring about the habit. They were quitting because the system made partial effort feel worthless. The intent was still there, but the system could not capture it.

Second, the logging burden was underestimated across every tool. Users described opening their tracker as a separate task rather than a natural extension of the habit itself, which meant on low-energy days, it was one more step to add.

Interview Insights

"I had a 47-day running streak and missed one day because I was sick. Seeing it reset to zero made me not want to open the app for a week. I knew logically I hadn't failed but the app said otherwise." – 22-year-old, had used three different habit trackers in the past year

"I do log my habits, but only on good days. On bad days I just don't open the app because I know I won't have anything to show for it. So my data only reflects my best days, not my real patterns." – 21-year-old student, primary habits: studying and exercise

"The problem with Notion is I spent more time designing my tracker than actually using it. By the time I had it looking perfect I had no energy left to log anything." – 19-year-old, switched between Notion and paper notebooks

"I did a 20-minute walk instead of my usual 5k and Strava counted it the same. That felt fine. But my habit tracker marked it as a missed run. Those are two different products saying two different things about the same day." – 22-year-old, primary habit: running

"What I actually want is something that says 'you showed up today, even if it wasn't your best,' not something that punishes me for being human." – 19-year-old, tried Habitica, abandoned after 3 weeks

3. Competitor Analysis

Competitor insights were drawn from hands-on testing, user interviews, App Store reviews, and qualitative research across r/habitica, r/getdisciplined, and r/productivity. Reddit surfaced recurring complaints at scale – streak resets, logging fatigue, and gamification burnout appeared across hundreds of threads unprompted, giving the analysis external validity beyond personal experience.

Habitica: Gamified habit tracker	
Pros	Cons
Highly engaging gamified reward system	Steep learning curve for non-gamers
Large community and social accountability features	Time spent managing the game can exceed time spent building habits
RPG-style progression makes habits feel like achievements	Overwhelming interface with too many game mechanics
Web + mobile availability	Losing health for missed habits can increase anxiety
Motivation for users who enjoy gaming	Social features can lead to negative pressure
	Requires constant internet connection
	Engagement drops once players reach higher levels
	Bugs and performance issues reported

Google Tasks: Simple task manager	
Pros	Cons
Easy to use and lightweight	Designed for tasks lists, not habits
Seamlessly integrated across the Google ecosystem	Poor support for habit repetition and streak tracking
Easy to create tasks using Google Assistant	Hard to view entire lists/habit systems at a glance
Clean UI focused on task completion	No habit analytics or progress tracking
Works well for basic reminders	Limited motivational elements

	“Feels more like a business productivity tool than personal growth”
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Notion: Highly customizable productivity workspace	
Pros	Cons
Highly customizable habit tracking system	Steep learning curve
Can integrate with goals, projects, calendars	Requires significant manual setup
Strong visual dashboards	Easy to overbuild overly complex systems
Multiple data view (table, board, calendar)	No habit reminders
Large collection of habit tracker templates	No built-in habit analytics or reporting

The competitor analysis pointed to a consistent gap: every existing tool optimizes for either engagement (Habitica) or flexibility (Notion) but none addresses the core behavioral failure: what happens when a user can't show up at full capacity. Habitica punishes absence. Notion requires manual effort that becomes a chore. Google Tasks ignores the habit dimension entirely. Elastic was designed specifically for the moment all three fail: **the low-energy day**.

Opportunity for my product after analyzing its competitors:

Habitica	Google Tasks	Notion
Minimal UI	Habit streak tracking	Pre-structured habit tracking
Low Cognitive Load	Habit analytics	Built-in habit analytics
Habit accountability without punishment	Long-term consistency visualization	Faster daily logging
Distraction-free environment	Habit completion times	Simplified dashboards

4. Defining the Real Problem

As I reflected on my experience and spoke with people who have used habit tracking tools, I realized the “lack of motivation” issue was actually a **mismatch between how habit systems are designed and how people really behave**.

Users typically set ambitious goals, assuming they will show up with the same energy every day. In reality, energy fluctuates due to work, health, and external demands. On days where users cannot meet their ideal version of a habit, existing trackers offer no flexibility. If the full task isn't completed, the effort feels meaningless, and users disengage.

This is reinforced by product design, as most habit trackers rely on binary completion and streaks. A missed day resets progress, creating a sense of failure that discourages users from continuing.

Core Problem: Habit systems break during low-energy periods, causing users to disengage despite continued intent.

Jobs To Be Done (JTBD)

To better understand what users actually need from a habit system, I reframed the problem using JTBD.

At a functional level, users want a way to **track progress toward long-term goals without adding extra effort**. However, at a deeper level, their need is behavioral and emotional.

Users want to **feel like they are making progress even on imperfect days**, rather than being punished for inconsistency. They want a system that **acknowledges partial effort**, so that showing up in any capacity still feels meaningful. Just as importantly, they want to maintain their identity as someone who is consistent, instead of being reminded of failure through broken streaks. Ultimately, users are trying to **build momentum that survives fluctuating energy**.

JTBD: When my energy fluctuates, I want to continue making progress toward my goals without feeling like I've failed, so that I can stay consistent over time.

This JTBD directly shaped my product decisions to prioritize flexibility, low friction, and recovery over rigid tracking and streak-based motivation.

Expanded Problem Statements

Based on this JBTD, I identified four key failure points in existing habit systems.

1. The **Binary Trap**: Partial effort is invisible to the system. When a user runs 10 minutes instead of 30, the tracker records a failure, which is exactly what it wasn't.
2. The **Maintenance Burden**: The act of tracking becomes the habit. When logging takes more effort than the habit itself, users stop logging, and eventually stop altogether.
3. The **Punishment Loop**: A single missed day erases visible progress. The system that was supposed to build consistency becomes the reason users quit.
4. The **Insight Gap**: Users log effort but never learn from it. Without feedback on whether their patterns are working, motivation becomes disconnected from behavior.

Prioritization & Rationale

My prioritization principle was to **focus on what causes users to quit first, then optimize what helps them improve.**

The Binary Trap and Maintenance Burden directly drive churn. In contrast, insights and advanced features improve the experience but do not address the root cause of disengagement. Hence, while all four problems are important, I focused on the ones that most directly impact **retention**.

I prioritized solving:

- The **Binary Trap** first, since unrecognized effort leads to immediate disengagement
- The **Maintenance Burden** second, since friction prevents sustained use
- The **Punishment Loop** third, to reduce drop-off after failure

I deprioritized the **Insight Gap** initially, since insights require consistent data to be useful.

Resulting Focus

This led to a clear product direction:

- Enable partial wins to sustain momentum
- Reduce logging friction to near zero
- Replace punitive feedback loops with more resilient mechanisms

5. Product Thesis

Habit systems should adapt to user energy rather than demand consistency from it.

This shifts success from perfect completion to **continuous progress**. Instead of asking the user “Did you complete the task?”, the system asks “Did you move forward today?”

To support this, I focused on three principles:

- **Elastic Effort:** scale based on energy
- **Frictionless Tracking:** no extra work
- **Progress-Based Feedback:** not punishment

These principles guided every decision I made.

6. Product Decisions & System Design

Each product decision maps directly to one of the four failure points. The goal was to ensure no design choice was decorative, every feature exists because a specific behavior was breaking.

1. The biggest shift was replacing binary tracking with an **Elastic Tier System**. This system ensures that effort is never lost. Even low-energy days contribute to progress, preserving momentum and

reinforcing consistency. Instead of marking a habit as simply “done” or “missed,” each action is classified based on effort:

- **Mini** represents the lowest viable version of a habit (e.g., reading one page or meditating for 3 minutes). It is designed to be “too easy to fail,” ensuring users can always show up, even on low-energy days
- **Plus** represents the standard level of effort expected on a typical day
- **Elite** captures high-effort sessions, for users to push themselves on high-energy days

2. To address the **Maintenance Burden**:

- a. I designed the system to minimize manual input wherever possible. For knowledge-based habits like coding or studying online, Elastic uses **passive tracking through a browser extension**, logging time spent on active tabs while filtering out idle activity.
- b. I integrated a **built-in Pomodoro focus timer** directly within the product. Users can attach a timer to a habit or goal, allowing them to track effort without switching to another app or tab.

This decision came from a simple observation: **every tab switch causes friction and increases distraction**. By keeping timing, tracking, and logging within the same environment, the system supports uninterrupted focus.

Solving the Binary Trap and Maintenance Burden keeps users logging. But without addressing what happens after a missed day, the system still breaks. That's what the fuel model solves.

3. To replace the Punishment Loop, I introduced a fuel-based system instead of streaks. Rather than resetting progress to zero after a missed day, consistency is measured as a “fuel level” that gradually increases with effort and slowly reduces during inactivity.

This creates a more forgiving model with continuous progress. Users are encouraged to recover and continue instead of restarting.

4. To ensure **long-term progress**, I introduced an **achievements system** that connects daily goals into milestones, as they are user-defined and tied to objectives. Users create them by setting specific targets linked to a goal, and progress is visualized through a gradual transition from grayscale to color and glow as they move further to completion.

This adds momentum within the users and anticipation, ensuring that effort turns into visual, long-term outcomes.

5. To bridge the **Insight Gap**, I introduced an **AI Mentor** that converts daily activity into actionable feedback. It generates a “Protocol Insight” summary that identifies patterns (e.g., over-reliance on Mini or stagnation at Plus) and recommends specific adjustments to improve progress.

This is visualized with a COM-B dashboard that calculates behavior across three dimensions:

- **Capability**: based on energy levels (Fuel Gauge)
- **Opportunity**: based on frequency of “Hide for Today” (external constraints)
- **Motivation**: based on effort quality (Mini vs Plus/Elite)

Each is scored out of five, immediately clarifying what's limiting users' progress, and what to adjust next.

All of these elements are connected through a **continuous feedback loop**. When a user logs an activity—either manually, through the timer, or passively—it is categorized into an effort tier, converted into progress, and reflected in their overall momentum.

7. Measuring Success

North Star: Progress Velocity – the smoothed average of effort logged per day across all tiers. This is the number that tells us if Elastic is doing its job: not whether users are opening the app, but whether they are consistently moving forward regardless of energy level. All other metrics are in service of this one.

To evaluate whether Elastic effectively supports long-term consistency, I defined a set of metrics across three categories: **Key Metric, Secondary Metrics, and Guardrails**.

Metric Type	Metric Name	Definition	Why It Matters
Key Metric	Progress Velocity	Smoothed average of effort (points) logged per day	Captures whether users are consistently making long-term measured progress
Secondary	Consistency Rate	% of days with at least a Mini log	Measures if users continue showing up, even on low-energy days
Secondary	Recovery Rate	% of users who return after inactivity	Indicates whether the system reduces drop-off after failure
Secondary	Effort Distribution	Ratio of Mini vs Plus vs Elite logs	Helps understand whether users are sustaining or burning out
Guardrail	Time to Log	Average time taken to log an action	Ensures tracking remains frictionless and not a chore
Guardrail	Passive Tracking Accuracy	% of correctly logged active sessions vs idle time	Prevents inflated or misleading progress from passive tracking
Guardrail	Over-reliance on Mini	% of users logging only Mini efforts over long periods	If users stagnate at Mini, the system may be too forgiving

8. Omitted Ideas

A key part of the process was deciding what not to build. I deprioritized:

- Social Features such as sharing or competition
 - Although competition can encourage users to complete tasks, researching competitors like Habitica displayed that it can also introduce pressure and comparison. For many

users, this leads to intimidation or negative feedback from friends, ultimately increasing churn rate

- Heavy gamification layers
 - Despite gamification being engaging, it does not work for all users. It can add unnecessary complexity and friction. I wanted Elastic to feel simple and intentional to encourage smooth onboarding for a wide range of users
- A mobile-first experience
 - While most habit trackers are designed for mobile, I found that logging habits on phones often introduces distraction, as users passively switch to different apps when they pick up their phones. Since many high-effort habits like coding or studying happen on desktops, I prioritized a browser-first experience that keeps users in their work environment and reduces context switching
- Advanced customization early on
 - Too many options can distract users from the primary goal of taking action. Given the broad range of potential users, I focused on building a simple, opinionated system that requires near zero onboarding friction

Overall, these decisions allowed me to focus on solving the core problem of **helping users stay consistent**, before expanding into features that optimize for engagement or personalization.

9. What Didn't Work

The most useful product decisions came from removing things that seemed right but weren't.

One early version of the product focused only on **short-term habits without linking them to long-term objectives**. While users could log activities like running or studying, there was no clear connection that they were building to a larger goal. Hence, habits felt isolated, and progress lacked a clever outcome, making it harder to sustain motivation.

I also initially **treated all missed logs the same**, regardless of context. This meant that if users couldn't complete a habit due to external factors, such as illness or unexpected commitments, they were still penalized. This created unnecessary guilt for situations outside their control. To address this issue, I introduced a **"Hide for Today"** option for individual habits and a **global "Snooze" mode** to temporarily pause all goals. This allowed users to maintain their sense of consistency without being punished for real-life interruptions.

An earlier version celebrated every completed action with confetti, including Mini logs. In testing, this felt rewarding for the first few sessions, but users stopped distinguishing between minimal effort and meaningful progress. When everything is celebrated equally, nothing feels earned. Confetti was limited to Elite-level efforts only, which restored the sense of distinction between showing up and pushing forward. When everything is celebrated equally, there is a lower feel of distinction between minimal effort and meaningful progress. I refined this by limiting celebratory confetti to **Elite-level efforts**. This helped prevent over-rewarding, which can dilute meaningful progress.

These iterations reinforced a key principle: **systems should differentiate between ability, effort, and growth, rather than treating all outcomes the same.**

10. Final Product & Key Learnings

Elastic is designed as a system where **every action contributes to momentum, regardless of its intensity**. Users can log habits manually, through a built-in focus timer, or passively via browser tracking. Each action is categorized into Mini, Plus, or Elite effort, ensuring that even low-energy days are captured and reflected as meaningful progress.

Instead of relying on fragile streaks, the system uses a **fuel-based model** where consistency builds gradually and decays slowly. Features like “Hide for Today” and Snooze mode account for real-life interruptions, allowing users to pause without breaking their sense of progress. Over time, daily actions connect directly to long-term goals, making progress visible, flexible, and continuous.

This project fundamentally changed the way I think about product design and user behavior. Key learning insights include:

Reducing friction is often more effective than increasing motivation. The users who stayed consistent longest weren't the most motivated, they were the ones for whom logging required the least effort.

Products should adapt to real user behavior, not ideal behavior. Every interviewee described the same pattern: ambitious setup, consistent use for one to two weeks, then a low-energy day that the system couldn't accommodate, followed by disengagement.

How a product handles failure determines long-term retention more than how it handles success. The broken streak moment was cited in every interview as the point where abandonment felt justified.

Complexity should be sequenced, not front-loaded. The fuel system, tier logic, and AI Mentor only work because the core logging loop is simple enough to survive a bad day. Features built on a fragile foundation fail when the foundation does.

Ultimately, Elastic shifts habit tracking from a system of accountability to one of **adaptation and resilience**, where the goal is of **continuous forward movement** rather than perfect execution.

11. Next Steps

With the core system validating that **flexibility and low friction improve consistency**, the next phase would be to make the system more **adaptive and intelligent over time**.

Phase 2: Adaptive Intelligence The AI Mentor evolves from a passive summarizer to an active guide, suggesting milestones based on observed behavior rather than waiting for users to define them. If a user has been over-relying on Mini logs for two weeks, the system flags it and recommends a specific adjustment. This phase is gated on two conditions: passive tracking accuracy above 80%, and over-reliance on Mini below 40%. Without both, the system doesn't have enough reliable data to make recommendations worth trusting.

Phase 3: Frictionless Input Manual logging is replaced as the primary input method. Users speak to create goals, log completions, or ask for feedback. Passive integrations – GitHub activity, productivity

tools, health apps – capture effort automatically in the background. The gate for this phase is simple: Time to Log must be consistently below 10 seconds in Phase 2, confirming that the friction problem is solved before removing the manual option entirely.

Phase 4: Environmental Awareness The system moves beyond the browser. Wearable and health app integrations bring in sleep, movement, and activity data – allowing Elastic to adapt goals and send notifications based on how the user's body is actually doing, not just what they've logged. A user coming off poor sleep gets a different target than one coming off a rest day. This phase is gated on Phase 3 passive tracking being stable, ensuring the system isn't adapting to incomplete data.

Together, these steps evolve Elastic from a passive tracker to a **responsive system that adapts to user energy, learns from their behavior, and actively guides them toward sustainable consistency**.

12. Research & Influences

Foundational

- Atomic Habits (Clear, 2018) – the concept of minimum viable habits directly informed the Mini tier design
- BJ Fogg's Tiny Habits – validated the behavioral case for scaling effort rather than demanding consistency
- Strava's weekly goal model – identified as the closest existing approach to elastic effort, and the gap it still leaves around effort quality

Supporting Sources

- [Expect Failure if You Have Poor Feedback Loops](#) – reinforced the decision to build observability into the core loop rather than treating feedback as a secondary feature
- [HelpGuide](#) – Burnout Prevention and Recovery – informed the "Hide for Today" and Snooze design decisions; burnout literature consistently shows that forced rest without guilt is a retention mechanism, not a failure
- [SustainabilityDirectory](#) – The All-or-Nothing Trap (SustainabilityDirectory) – provided behavioral framing for why binary systems produce dropout rather than discipline
- [PrincetonUniversityPress](#) – Why Are Habits So Sticky? – grounded the fuel model in habit formation research; consistency of context matters more than consistency of effort